

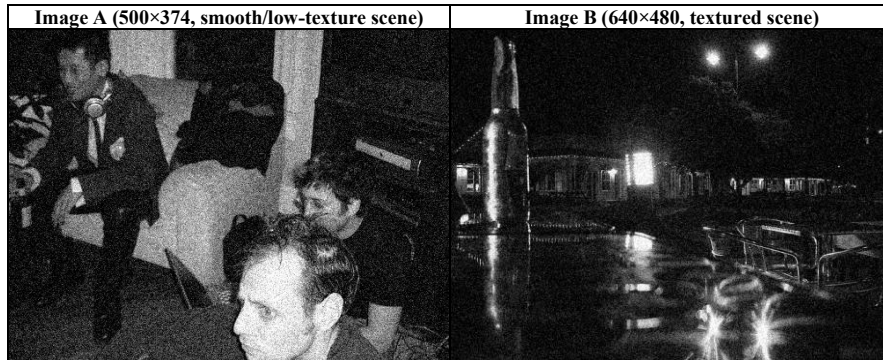
Image Processing Assignment 3

Name: Yuvicka

Roll No.: 1024240016

Group: 3X11

Task 1 — Restoration Filter Investigation



1. Arithmetic Mean Filter

Formula: $g(x,y) = (1/mn) \sum f(s,t)$

Noise Setting	Noisy — Image A	Filtered — Image A	Noisy — Image B	Filtered — Image B
Gaussian (mean=0, var=0.05)	 noisy	 PSNR 22.0 SSIM 0.38	 noisy	 PSNR 20.8 SSIM 0.31
Rayleigh (scale a=0.3)	 noisy	 PSNR 9.0 SSIM 0.27	 noisy	 PSNR 8.6 SSIM 0.16
Erlang (shape b=2, rate a=5)	 noisy	 PSNR 8.8 SSIM 0.24	 noisy	 PSNR 8.3 SSIM 0.14
Exponential (rate a=1)	 noisy	 PSNR 5.4 SSIM 0.17	 noisy	 PSNR 4.5 SSIM 0.07
Uniform (range [-0.3, 0.3])	 noisy	 PSNR 23.7 SSIM 0.46	 noisy	 PSNR 22.2 SSIM 0.38

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Salt-and-pepper (density d=0.15)				
	noisy	PSNR 20.2 SSIM 0.32	noisy	PSNR 19.1 SSIM 0.25

2. Geometric Mean Filter

Formula: $g(x,y) = [\prod f(s,t)]^{1/mn}$

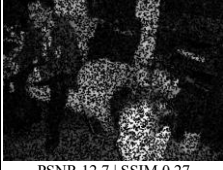
Noise Setting	Noisy — Image A	Filtered — Image A	Noisy — Image B	Filtered — Image B
Gaussian (mean=0, var=0.05)				
	noisy	PSNR 17.6 SSIM 0.19	noisy	PSNR 18.5 SSIM 0.23
Rayleigh (scale a=0.3)				
	noisy	PSNR 9.8 SSIM 0.28	noisy	PSNR 9.6 SSIM 0.17
Erlang (shape b=2, rate a=5)				
	noisy	PSNR 9.9 SSIM 0.27	noisy	PSNR 9.7 SSIM 0.16
Exponential (rate a=1)				
	noisy	PSNR 6.7 SSIM 0.18	noisy	PSNR 6.1 SSIM 0.08
Uniform (range [-0.3, 0.3])				
	noisy	PSNR 19.6 SSIM 0.25	noisy	PSNR 19.4 SSIM 0.25
Salt-and-pepper (density d=0.15)				
	noisy	PSNR 12.7 SSIM 0.27	noisy	PSNR 16.4 SSIM 0.41

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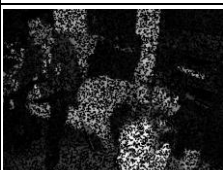
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3. Harmonic Mean Filter

Formula: $g(x,y) = mn / \sum (1/f(s,t))$

Noise Setting	Noisy — Image A	Filtered — Image A	Noisy — Image B	Filtered — Image B
Gaussian (mean=0, var=0.05)	 noisy	 PSNR 16.4 SSIM 0.14	 noisy	 PSNR 17.7 SSIM 0.19
Rayleigh (scale a=0.3)	 noisy	 PSNR 10.6 SSIM 0.29	 noisy	 PSNR 10.6 SSIM 0.17
Erlang (shape b=2, rate a=5)	 noisy	 PSNR 11.1 SSIM 0.28	 noisy	 PSNR 11.3 SSIM 0.17
Exponential (rate a=1)	 noisy	 PSNR 8.3 SSIM 0.19	 noisy	 PSNR 8.1 SSIM 0.09
Uniform (range [-0.3, 0.3])	 noisy	 PSNR 18.6 SSIM 0.22	 noisy	 PSNR 18.7 SSIM 0.22
Salt-and-pepper (density d=0.15)	 noisy	 PSNR 11.9 SSIM 0.25	 noisy	 PSNR 15.6 SSIM 0.37

4a. Contraharmonic Mean Filter (Q = 1, removes pepper noise)

Formula: $g(x,y) = \sum f(s,t)^{(Q+1)} / \sum f(s,t)^Q$, Q = 1

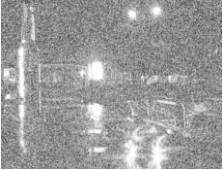
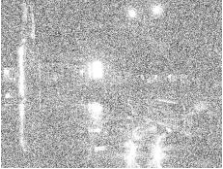
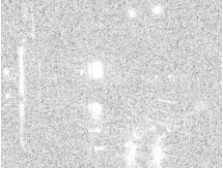
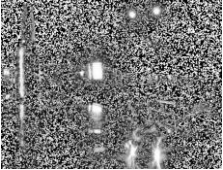
Noise Setting	Noisy — Image A	Filtered — Image A	Noisy — Image B	Filtered — Image B
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Gaussian (mean=0, var=0.05)	 noisy	 PSNR 13.9 SSIM 0.23	 noisy	 PSNR 12.6 SSIM 0.16
Rayleigh (scale a=0.3)	 noisy	 PSNR 7.6 SSIM 0.24	 noisy	 PSNR 7.0 SSIM 0.13
Erlang (shape b=2, rate a=5)	 noisy	 PSNR 6.8 SSIM 0.20	 noisy	 PSNR 6.0 SSIM 0.10
Exponential (rate a=1)	 noisy	 PSNR 3.7 SSIM 0.15	 noisy	 PSNR 2.7 SSIM 0.06
Uniform (range [-0.3, 0.3])	 noisy	 PSNR 16.5 SSIM 0.34	 noisy	 PSNR 15.1 SSIM 0.25
Salt-and-pepper (density d=0.15)	 noisy	 PSNR 8.3 SSIM 0.16	 noisy	 PSNR 6.0 SSIM 0.08

4b. Contraharmonic Mean Filter (Q = -1, removes salt noise)

Formula: $g(x,y) = \frac{\sum f(s,t)^{(Q+1)}}{\sum f(s,t)^Q}$, Q = -1

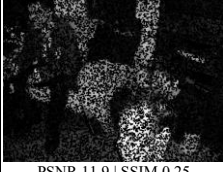
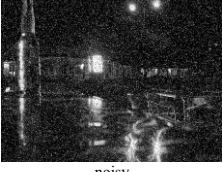
Noise Setting	Noisy — Image A	Filtered — Image A	Noisy — Image B	Filtered — Image B
Gaussian (mean=0, var=0.05)	 noisy	 PSNR 16.4 SSIM 0.14	 noisy	 PSNR 17.7 SSIM 0.19

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Rayleigh (scale a=0.3)	 noisy	 PSNR 10.6 SSIM 0.29	 noisy	 PSNR 10.6 SSIM 0.17
Erlang (shape b=2, rate a=5)	 noisy	 PSNR 11.1 SSIM 0.28	 noisy	 PSNR 11.3 SSIM 0.17
Exponential (rate a=1)	 noisy	 PSNR 8.3 SSIM 0.19	 noisy	 PSNR 8.1 SSIM 0.09
Uniform (range [-0.3, 0.3])	 noisy	 PSNR 18.6 SSIM 0.22	 noisy	 PSNR 18.7 SSIM 0.22
Salt-and-pepper (density d=0.15)	 noisy	 PSNR 11.9 SSIM 0.25	 noisy	 PSNR 15.6 SSIM 0.37

5. Median Filter

Formula: $g(x,y) = \text{median} \{ f(s,t) \}$

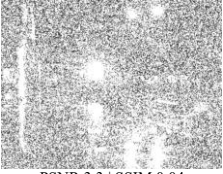
Noise Setting	Noisy — Image A	Filtered — Image A	Noisy — Image B	Filtered — Image B
Gaussian (mean=0, var=0.05)	 noisy	 PSNR 21.4 SSIM 0.32	 noisy	 PSNR 21.2 SSIM 0.29
Rayleigh (scale a=0.3)	 noisy	 PSNR 9.0 SSIM 0.23	 noisy	 PSNR 8.8 SSIM 0.14

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Erlang (shape b=2, rate a=5)	 noisy	 PSNR 9.1 SSIM 0.20	 noisy	 PSNR 8.9 SSIM 0.12
Exponential (rate a=1)	 noisy	 PSNR 4.0 SSIM 0.12	 noisy	 PSNR 3.3 SSIM 0.04
Uniform (range [-0.3, 0.3])	 noisy	 PSNR 21.4 SSIM 0.32	 noisy	 PSNR 21.3 SSIM 0.29
Salt-and-pepper (density d=0.15)	 noisy	 PSNR 30.0 SSIM 0.90	 noisy	 PSNR 26.2 SSIM 0.86

6. Min Filter

Formula: $g(x,y) = \min \{ f(s,t) \}$

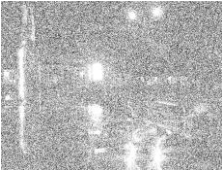
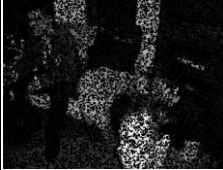
Noise Setting	Noisy — Image A	Filtered — Image A	Noisy — Image B	Filtered — Image B
Gaussian (mean=0, var=0.05)	 noisy	 PSNR 13.2 SSIM 0.09	 noisy	 PSNR 16.0 SSIM 0.16
Rayleigh (scale a=0.3)	 noisy	 PSNR 17.3 SSIM 0.32	 noisy	 PSNR 16.8 SSIM 0.22
Erlang (shape b=2, rate a=5)	 noisy	 PSNR 18.8 SSIM 0.35	 noisy	 PSNR 18.3 SSIM 0.25

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Exponential (rate a=1)	 noisy	 PSNR 16.2 SSIM 0.23	 noisy	 PSNR 15.8 SSIM 0.16
Uniform (range [-0.3, 0.3])	 noisy	 PSNR 15.0 SSIM 0.16	 noisy	 PSNR 16.8 SSIM 0.18
Salt-and-pepper (density d=0.15)	 noisy	 PSNR 11.8 SSIM 0.23	 noisy	 PSNR 15.4 SSIM 0.32

7. Max Filter

Formula: $g(x,y) = \max \{ f(s,t) \}$

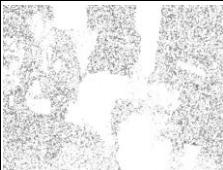
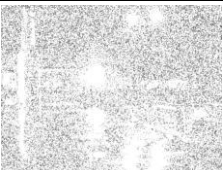
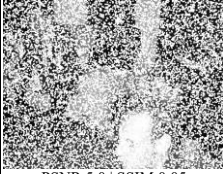
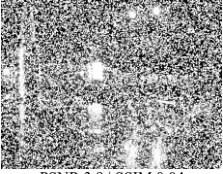
Noise Setting	Noisy — Image A	Filtered — Image A	Noisy — Image B	Filtered — Image B
Gaussian (mean=0, var=0.05)	 noisy	 PSNR 9.2 SSIM 0.16	 noisy	 PSNR 8.8 SSIM 0.10
Rayleigh (scale a=0.3)	 noisy	 PSNR 4.0 SSIM 0.16	 noisy	 PSNR 3.3 SSIM 0.05
Erlang (shape b=2, rate a=5)	 noisy	 PSNR 3.0 SSIM 0.14	 noisy	 PSNR 2.2 SSIM 0.04
Exponential (rate a=1)	 noisy	 PSNR 2.0 SSIM 0.18	 noisy	 PSNR 1.0 SSIM 0.06

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Uniform (range [-0.3, 0.3])	 noisy	 PSNR 11.9 SSIM 0.29	 noisy	 PSNR 11.5 SSIM 0.19
Salt-and-pepper (density d=0.15)	 noisy	 PSNR 5.0 SSIM 0.05	 noisy	 PSNR 3.9 SSIM 0.04

8. Midpoint Filter

Formula: $g(x,y) = 1/2 [\max f(s,t) + \min f(s,t)]$

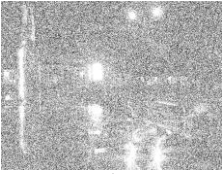
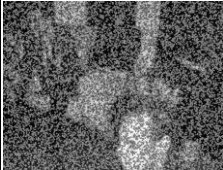
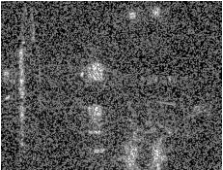
Noise Setting	Noisy — Image A	Filtered — Image A	Noisy — Image B	Filtered — Image B
Gaussian (mean=0, var=0.05)	 noisy	 PSNR 17.6 SSIM 0.26	 noisy	 PSNR 16.2 SSIM 0.20
Rayleigh (scale a=0.3)	 noisy	 PSNR 8.6 SSIM 0.25	 noisy	 PSNR 7.9 SSIM 0.13
Erlang (shape b=2, rate a=5)	 noisy	 PSNR 8.0 SSIM 0.24	 noisy	 PSNR 7.2 SSIM 0.12
Exponential (rate a=1)	 noisy	 PSNR 6.9 SSIM 0.21	 noisy	 PSNR 6.0 SSIM 0.12
Uniform (range [-0.3, 0.3])	 noisy	 PSNR 21.5 SSIM 0.50	 noisy	 PSNR 19.4 SSIM 0.37

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Salt-and-pepper (density d=0.15)				
	noisy	PSNR 11.1 SSIM 0.07	noisy	PSNR 10.2 SSIM 0.05

9a. Alpha-Trimmed Mean Filter (d = 2)

Formula: $g(x,y) = 1/(mn-d) \sum \text{remaining pixels after trimming } d/2 \text{ highest} + d/2 \text{ lowest, } d = 2$

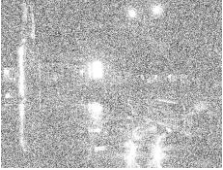
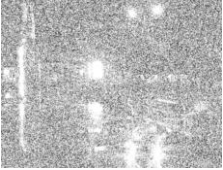
Noise Setting	Noisy — Image A	Filtered — Image A	Noisy — Image B	Filtered — Image B
Gaussian (mean=0, var=0.05)				
	noisy	PSNR 22.6 SSIM 0.38	noisy	PSNR 21.7 SSIM 0.32
Rayleigh (scale a=0.3)				
	noisy	PSNR 9.1 SSIM 0.26	noisy	PSNR 8.7 SSIM 0.15
Erlang (shape b=2, rate a=5)				
	noisy	PSNR 9.0 SSIM 0.23	noisy	PSNR 8.6 SSIM 0.13
Exponential (rate a=1)				
	noisy	PSNR 4.9 SSIM 0.15	noisy	PSNR 4.1 SSIM 0.06
Uniform (range [-0.3, 0.3])				
	noisy	PSNR 23.5 SSIM 0.42	noisy	PSNR 22.4 SSIM 0.36
Salt-and-pepper (density d=0.15)				
	noisy	PSNR 23.5 SSIM 0.50	noisy	PSNR 22.2 SSIM 0.44

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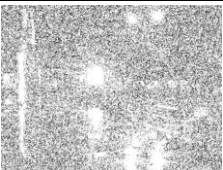
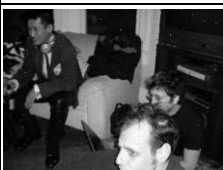
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9b. Alpha-Trimmed Mean Filter (d = 6)

Formula: $g(x,y) = 1/(mn-d) \sum \text{remaining pixels after trimming } d/2 \text{ highest} + d/2 \text{ lowest, } d = 6$

Noise Setting	Noisy — Image A	Filtered — Image A	Noisy — Image B	Filtered — Image B
Gaussian (mean=0, var=0.05)	 noisy	 PSNR 22.1 SSIM 0.35	 noisy	 PSNR 21.8 SSIM 0.32
Rayleigh (scale a=0.3)	 noisy	 PSNR 9.1 SSIM 0.24	 noisy	 PSNR 8.8 SSIM 0.15
Erlang (shape b=2, rate a=5)	 noisy	 PSNR 9.2 SSIM 0.21	 noisy	 PSNR 8.9 SSIM 0.13
Exponential (rate a=1)	 noisy	 PSNR 4.3 SSIM 0.13	 noisy	 PSNR 3.5 SSIM 0.04
Uniform (range [-0.3, 0.3])	 noisy	 PSNR 22.3 SSIM 0.36	 noisy	 PSNR 22.0 SSIM 0.32
Salt-and-pepper (density d=0.15)	 noisy	 PSNR 29.5 SSIM 0.87	 noisy	 PSNR 26.1 SSIM 0.82

10. Adaptive Local Noise Reduction Filter

Formula: $g(x,y) = z - (\sigma^2_{\text{noise}} / \sigma^2_{\text{local}}) \cdot (z - m_{\text{local}})$, ratio capped at 1

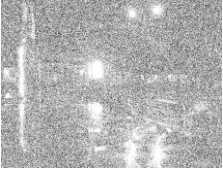
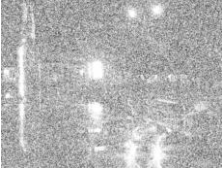
Noise Setting	Noisy — Image A	Filtered — Image A	Noisy — Image B	Filtered — Image B
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Gaussian (mean=0, var=0.05)	 noisy	 PSNR 19.3 SSIM 0.28	 noisy	 PSNR 19.3 SSIM 0.25
Rayleigh (scale a=0.3)	 noisy	 PSNR 9.0 SSIM 0.26	 noisy	 PSNR 8.6 SSIM 0.16
Erlang (shape b=2, rate a=5)	 noisy	 PSNR 8.8 SSIM 0.24	 noisy	 PSNR 8.3 SSIM 0.14
Exponential (rate a=1)	 noisy	 PSNR 5.4 SSIM 0.17	 noisy	 PSNR 4.5 SSIM 0.07
Uniform (range [-0.3, 0.3])	 noisy	 PSNR 21.1 SSIM 0.35	 noisy	 PSNR 21.0 SSIM 0.31
Salt-and-pepper (density d=0.15)	 noisy	 PSNR 17.6 SSIM 0.26	 noisy	 PSNR 17.0 SSIM 0.22

11. Adaptive Median Filter ($S_{max} = 7$)

Formula: Window grows $3 \rightarrow 5 \rightarrow 7$ per pixel until level-A conditions ($z_{min} < z_{med} < z_{max}$) are satisfied

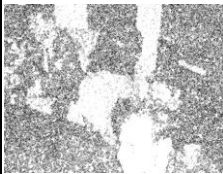
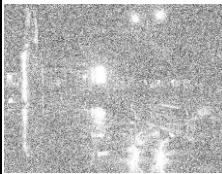
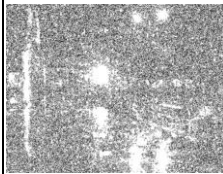
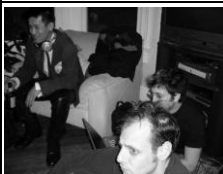
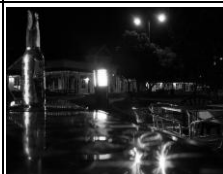
Noise Setting	Noisy — Image A	Filtered — Image A	Noisy — Image B	Filtered — Image B
Gaussian (mean=0, var=0.05)	 noisy	 PSNR 17.7 SSIM 0.19	 noisy	 PSNR 18.0 SSIM 0.18

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Rayleigh (scale $a=0.3$)	 noisy	 PSNR 8.8 SSIM 0.17	 noisy	 PSNR 8.4 SSIM 0.10
Erlang (shape $b=2$, rate $a=5$)	 noisy	 PSNR 8.7 SSIM 0.15	 noisy	 PSNR 8.2 SSIM 0.09
Exponential (rate $a=1$)	 noisy	 PSNR 5.2 SSIM 0.11	 noisy	 PSNR 4.5 SSIM 0.04
Uniform (range $[-0.3, 0.3]$)	 noisy	 PSNR 18.5 SSIM 0.21	 noisy	 PSNR 18.8 SSIM 0.20
Salt-and-pepper (density $d=0.15$)	 noisy	 PSNR 35.3 SSIM 0.96	 noisy	 PSNR 30.1 SSIM 0.94

Question 1 — Why do two visually similar noisy patches need different filters?

Noise types have different statistics, so the filter must match the noise model. **Median filtering** is highly effective for salt-and-pepper noise because it removes extreme outliers, but it offers little benefit for Gaussian noise. In contrast, **mean-based filters** can blur images and spread salt-and-pepper corruption because they average extreme values instead of rejecting them.

In real applications such as medical imaging, choosing the wrong filter can blur important details while failing to remove the actual noise, potentially affecting diagnosis.

Task 2 — Filter-vs-Noise Suitability Analysis

☐ **Gaussian noise:** Arithmetic mean performs better than geometric mean because it is a better fit for zero-mean, symmetric Gaussian noise.

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□ **Contraharmonic filter:** The sign of **Q** matters. **Q > 0** removes pepper noise, while **Q < 0** removes salt noise. Using the wrong sign can worsen the corruption.

□ **Min vs. Max filter:** The **min filter** is better for salt noise, while the **max filter** is better for pepper noise. For mixed salt-and-pepper noise, the wrong filter can spread the corruption.

□ **Adaptive vs. fixed median:** At high noise density, the **adaptive median** performs better than a fixed 3×3 median because it increases the window size only when necessary, reducing noise while preserving details.

Task 3 — Quantitative Summary and Comparison

Consolidated PSNR and MSE tables, one pair per noise family, rows = filter (all variants and window sizes), columns = image x severity setting:

Gaussian noise — PSNR (dB), higher is better

Filter	A: gaussian 0 0.01	A: gaussian 0 0.05	B: gaussian 0 0.01	B: gaussian 0 0.05
adaptive local noise reduction k3	25.77	19.25	25.11	19.34
adaptive local noise reduction k5	26.23	19.98	25.39	20.01
adaptive median Smax7	23.81	17.69	23.65	17.99
alpha trimmed k3 d2	27.86	22.58	25.50	21.70
alpha trimmed k3 d6	27.30	22.11	25.38	21.76
alpha trimmed k5 d2	27.22	23.65	24.29	21.68
alpha trimmed k5 d6	27.54	24.42	24.48	22.47
arithmetic mean k3	27.70	21.98	25.16	20.76
arithmetic mean k5	26.95	22.98	24.10	21.03
contraharmonic k3 Q-1	21.46	16.35	20.46	17.69
contraharmonic k3 Q1	21.56	13.90	19.17	12.57
contraharmonic k5 Q-1	18.71	14.50	18.58	16.24
contraharmonic k5 Q1	20.60	13.63	18.31	12.23
geometric mean k3	22.63	17.61	21.40	18.48
geometric mean k5	20.80	16.69	19.99	17.53
harmonic mean k3	21.46	16.35	20.46	17.69
harmonic mean k5	18.71	14.50	18.58	16.24
max k3	15.30	9.15	14.49	8.82
max k5	12.73	7.06	11.83	6.66
median k3	26.69	21.38	24.97	21.18
median k5	27.45	23.91	24.30	22.58
midpoint k3	24.50	17.58	22.13	16.23
midpoint k5	21.83	15.34	19.56	13.94
min k3	17.35	13.20	18.22	16.01
min k5	15.06	11.53	16.47	14.71

Gaussian noise — MSE, lower is better

Filter	A: gaussian 0 0.01	A: gaussian 0 0.05	B: gaussian 0 0.01	B: gaussian 0 0.05
adaptive local noise reduction k3	172.09	772.52	200.47	757.45
adaptive local noise reduction k5	154.80	653.05	187.79	649.19
adaptive median Smax7	270.39	1107.09	280.65	1033.37
alpha trimmed k3 d2	106.44	359.12	183.21	440.08
alpha trimmed k3 d6	121.10	399.94	188.53	433.72
alpha trimmed k5 d2	123.41	280.41	241.92	441.81
alpha trimmed k5 d6	114.64	235.17	231.83	367.87
arithmetic mean k3	110.42	412.25	198.18	546.15
arithmetic mean k5	131.35	327.50	253.19	513.28
contraharmonic k3 Q-1	464.83	1506.74	584.95	1107.66
contraharmonic k3 Q1	454.30	2649.54	787.07	3599.39
contraharmonic k5 Q-1	875.64	2308.46	902.72	1546.81
contraharmonic k5 Q1	566.14	2817.44	959.66	3895.19
geometric mean k3	354.75	1127.70	470.54	922.81
geometric mean k5	541.27	1392.20	652.45	1148.87
harmonic mean k3	464.83	1506.74	584.95	1107.66
harmonic mean k5	875.64	2308.46	902.72	1546.81
max k3	1920.18	7901.25	2314.19	8531.63
max k5	3466.16	12801.91	4271.47	14018.56
median k3	139.30	473.18	207.01	496.09
median k5	117.02	264.51	241.37	358.91
midpoint k3	230.52	1136.04	397.97	1550.02

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midpoint k5	426.71	1900.61	720.18	2624.18
min k3	1196.38	3114.48	979.28	1627.88
min k5	2029.23	4567.00	1465.69	2196.17

Rayleigh noise — PSNR (dB), higher is better

Filter	A: rayleigh 0.1	A: rayleigh 0.3	B: rayleigh 0.1	B: rayleigh 0.3
adaptive local noise reduction k3	18.12	8.96	18.04	8.56
adaptive local noise reduction k5	18.16	9.03	18.09	8.63
adaptive median Smax7	17.88	8.78	17.70	8.40
alpha trimmed k3 d2	18.29	9.06	17.94	8.74
alpha trimmed k3 d6	18.39	9.09	18.09	8.84
alpha trimmed k5 d2	18.08	9.08	17.61	8.72
alpha trimmed k5 d6	18.23	9.14	17.78	8.84
arithmetic mean k3	18.11	8.99	17.73	8.59
arithmetic mean k5	17.96	9.03	17.48	8.63
contraharmonic k3 Q-1	19.46	10.63	19.33	10.61
contraharmonic k3 Q1	16.81	7.62	16.03	6.99
contraharmonic k5 Q-1	19.52	10.97	19.25	11.02
contraharmonic k5 Q1	16.37	7.53	15.59	6.90
geometric mean k3	18.80	9.78	18.59	9.55
geometric mean k5	18.78	9.93	18.41	9.71
harmonic mean k3	19.46	10.63	19.33	10.61
harmonic mean k5	19.52	10.97	19.25	11.02
max k3	12.07	3.97	11.60	3.28
max k5	10.35	2.95	9.78	2.15
median k3	18.28	9.04	17.99	8.80
median k5	18.29	9.14	17.87	8.95
midpoint k3	17.24	8.56	16.73	7.93
midpoint k5	16.36	8.30	15.66	7.50
min k3	24.42	17.25	22.31	16.80
min k5	21.65	20.41	20.83	19.56

Rayleigh noise — MSE, lower is better

Filter	A: rayleigh 0.1	A: rayleigh 0.3	B: rayleigh 0.1	B: rayleigh 0.3
adaptive local noise reduction k3	1002.00	8265.30	1021.59	9069.44
adaptive local noise reduction k5	992.20	8132.09	1010.47	8907.38
adaptive median Smax7	1058.91	8620.44	1104.93	9407.43
alpha trimmed k3 d2	963.79	8066.14	1045.36	8695.61
alpha trimmed k3 d6	942.98	8015.59	1009.71	8496.19
alpha trimmed k5 d2	1011.08	8035.26	1126.54	8729.38
alpha trimmed k5 d6	977.59	7921.45	1083.51	8493.33
arithmetic mean k3	1003.81	8207.54	1097.30	9002.67
arithmetic mean k5	1039.56	8123.90	1162.56	8915.81
contraharmonic k3 Q-1	736.93	5625.20	757.88	5649.54
contraharmonic k3 Q1	1355.23	11251.39	1621.75	13000.30
contraharmonic k5 Q-1	726.77	5200.97	772.37	5137.04
contraharmonic k5 Q1	1498.84	11477.52	1796.71	13272.02
geometric mean k3	857.47	6839.41	900.57	7210.74
geometric mean k5	861.08	6605.35	937.49	6951.64
harmonic mean k3	736.93	5625.20	757.88	5649.54
harmonic mean k5	726.77	5200.97	772.37	5137.04
max k3	4032.88	26045.40	4498.47	30519.88
max k5	5999.92	32929.46	6842.03	39596.80
median k3	966.51	8108.41	1033.59	8573.97
median k5	963.40	7927.91	1061.86	8285.48
midpoint k3	1227.18	9064.51	1380.98	10473.73
midpoint k5	1501.73	9617.47	1765.77	11558.34
min k3	235.24	1223.84	382.32	1359.09
min k5	444.27	591.28	536.79	719.30

Erlang noise — PSNR (dB), higher is better

Filter	A: erlang 2 10	A: erlang 2 5	B: erlang 2 10	B: erlang 2 5
adaptive local noise reduction k3	13.99	8.75	13.76	8.28
adaptive local noise reduction k5	14.15	8.86	13.93	8.39
adaptive median Smax7	14.06	8.71	13.85	8.24
alpha trimmed k3 d2	14.60	8.96	14.37	8.59
alpha trimmed k3 d6	14.94	9.16	14.76	8.91

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alpha trimmed k5 d2	14.45	8.94	14.16	8.53
alpha trimmed k5 d6	14.78	9.12	14.53	8.81
arithmetic mean k3	14.14	8.79	13.84	8.32
arithmetic mean k5	14.18	8.86	13.85	8.40
contraharmonic k3 Q-1	16.36	11.13	16.63	11.26
contraharmonic k3 Q1	12.00	6.84	11.27	6.04
contraharmonic k5 Q-1	16.72	11.59	16.97	11.84
contraharmonic k5 Q1	11.83	6.75	11.09	5.94
geometric mean k3	15.27	9.93	15.27	9.74
geometric mean k5	15.46	10.15	15.40	9.98
harmonic mean k3	16.36	11.13	16.63	11.26
harmonic mean k5	16.72	11.59	16.97	11.84
max k3	7.02	3.00	6.55	2.16
max k5	5.30	2.20	4.70	1.23
median k3	14.90	9.13	14.72	8.90
median k5	15.15	9.38	14.92	9.23
midpoint k3	12.38	7.99	11.89	7.22
midpoint k5	11.25	7.79	10.57	6.84
min k3	23.57	18.78	21.83	18.28
min k5	21.94	21.49	20.99	20.49

Erlang noise — MSE, lower is better

Filter	A: erlang 2 10	A: erlang 2 5	B: erlang 2 10	B: erlang 2 5
adaptive local noise reduction k3	2596.12	8663.87	2733.89	9670.45
adaptive local noise reduction k5	2502.76	8461.05	2628.26	9417.41
adaptive median Smax7	2554.74	8760.01	2679.66	9748.45
alpha trimmed k3 d2	2254.54	8267.01	2375.80	8992.95
alpha trimmed k3 d6	2082.95	7893.74	2173.50	8349.64
alpha trimmed k5 d2	2334.34	8298.43	2494.27	9115.13
alpha trimmed k5 d6	2164.56	7959.56	2291.66	8558.19
arithmetic mean k3	2507.78	8600.58	2684.08	9580.29
arithmetic mean k5	2482.86	8446.88	2680.33	9398.01
contraharmonic k3 Q-1	1504.46	5016.50	1413.13	4867.87
contraharmonic k3 Q1	4103.02	13461.86	4851.72	16196.21
contraharmonic k5 Q-1	1384.96	4511.98	1306.85	4255.86
contraharmonic k5 Q1	4265.52	13738.42	5063.08	16568.71
geometric mean k3	1931.68	6605.82	1932.26	6908.91
geometric mean k5	1850.67	6280.89	1874.48	6526.22
harmonic mean k3	1504.46	5016.50	1413.13	4867.87
harmonic mean k5	1384.96	4511.98	1306.85	4255.86
max k3	12909.95	32581.24	14407.21	39544.93
max k5	19211.42	39219.22	22024.08	48968.30
median k3	2104.34	7952.77	2194.60	8380.16
median k5	1986.71	7501.17	2093.16	7757.35
midpoint k3	3759.48	10326.13	4204.06	12322.07
midpoint k5	4878.57	10813.18	5697.04	13470.81
min k3	285.58	860.52	427.04	966.42
min k5	415.64	461.34	517.55	580.85

Exponential noise — PSNR (dB), higher is better

Filter	A: exponential 1	A: exponential 3	B: exponential 1	B: exponential 3
adaptive local noise reduction k3	5.37	10.42	4.53	9.94
adaptive local noise reduction k5	5.45	10.62	4.62	10.15
adaptive median Smax7	5.18	10.47	4.51	9.96
alpha trimmed k3 d2	4.95	10.92	4.14	10.57
alpha trimmed k3 d6	4.26	11.46	3.48	11.25
alpha trimmed k5 d2	5.28	10.86	4.46	10.44
alpha trimmed k5 d6	4.94	11.30	4.14	11.00
arithmetic mean k3	5.38	10.49	4.54	10.01
arithmetic mean k5	5.46	10.65	4.62	10.17
contraharmonic k3 Q-1	8.33	14.49	8.13	15.06
contraharmonic k3 Q1	3.72	7.30	2.67	6.34
contraharmonic k5 Q-1	9.25	15.26	9.46	16.03
contraharmonic k5 Q1	3.67	7.18	2.61	6.20
geometric mean k3	6.68	12.50	6.12	12.52
geometric mean k5	6.99	12.91	6.50	12.99
harmonic mean k3	8.33	14.49	8.13	15.06
harmonic mean k5	9.25	15.26	9.46	16.03
max k3	2.00	3.10	0.97	2.27
max k5	1.99	2.18	0.96	1.21
median k3	4.03	11.44	3.25	11.26

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median k5	3.91	12.10	3.20	11.95
midpoint k3	6.93	8.80	5.98	7.98
midpoint k5	7.73	8.29	6.69	7.26
min k3	16.22	23.74	15.76	21.97
min k5	21.68	21.93	20.65	20.93

Exponential noise — MSE, lower is better

Filter	A: exponential 1	A: exponential 3	B: exponential 1	B: exponential 3
adaptive local noise reduction k3	18869.03	5901.56	22888.14	6591.81
adaptive local noise reduction k5	18535.46	5641.59	22443.93	6283.41
adaptive median Smax7	19722.21	5836.95	23033.34	6556.63
alpha trimmed k3 d2	20802.98	5259.85	25092.33	5704.88
alpha trimmed k3 d6	24379.03	4641.35	29183.21	4873.64
alpha trimmed k5 d2	19295.74	5340.10	23295.55	5874.43
alpha trimmed k5 d6	20867.31	4821.18	25051.81	5160.91
arithmetic mean k3	18829.18	5803.63	22843.73	6488.86
arithmetic mean k5	18514.37	5597.89	22423.18	6258.48
contraharmonic k3 Q-1	9553.26	2311.94	10005.60	2028.70
contraharmonic k3 Q1	27636.01	12098.49	35168.35	15111.97
contraharmonic k5 Q-1	7729.27	1936.92	7366.63	1620.97
contraharmonic k5 Q1	27961.91	12445.66	35621.39	15601.29
geometric mean k3	13951.29	3656.28	15897.13	3635.80
geometric mean k5	13016.86	3328.35	14560.53	3268.05
harmonic mean k3	9553.26	2311.94	10005.60	2028.70
harmonic mean k5	7729.27	1936.92	7366.63	1620.97
max k3	40992.93	31861.09	51984.94	38593.75
max k5	41096.10	39324.81	52145.71	49169.50
median k3	25691.43	4663.77	30742.90	4869.96
median k5	26443.89	4007.72	31155.94	4146.01
midpoint k3	13170.25	8566.48	16402.78	10361.52
midpoint k5	10975.60	9648.11	13933.52	12210.22
min k3	1552.65	274.93	1727.89	412.71
min k5	441.82	417.16	560.24	524.95

Uniform noise — PSNR (dB), higher is better

Filter	A: uniform -0.1 0.1	A: uniform -0.3 0.3	B: uniform -0.1 0.1	B: uniform -0.3 0.3
adaptive local noise reduction k3	30.18	21.08	28.84	21.01
adaptive local noise reduction k5	30.22	21.44	28.80	21.30
adaptive median Smax7	26.87	18.47	26.17	18.82
alpha trimmed k3 d2	29.95	23.49	26.55	22.45
alpha trimmed k3 d6	29.32	22.30	26.42	21.99
alpha trimmed k5 d2	28.03	24.57	24.75	22.37
alpha trimmed k5 d6	28.32	24.90	24.81	22.80
arithmetic mean k3	30.00	23.66	26.43	22.16
arithmetic mean k5	27.83	24.29	24.68	22.07
contraharmonic k3 Q-1	24.29	18.63	22.09	18.68
contraharmonic k3 Q1	25.65	16.52	22.30	15.05
contraharmonic k5 Q-1	20.93	16.85	20.09	17.31
contraharmonic k5 Q1	23.46	16.14	20.78	14.63
geometric mean k3	25.61	19.64	23.14	19.45
geometric mean k5	23.02	18.40	21.38	18.37
harmonic mean k3	24.29	18.63	22.09	18.68
harmonic mean k5	20.93	16.85	20.09	17.31
max k3	19.11	11.95	17.50	11.55
max k5	16.15	10.32	14.44	9.83
median k3	28.71	21.44	26.07	21.31
median k5	28.48	23.65	24.57	22.54
midpoint k3	28.22	21.52	24.62	19.41
midpoint k5	24.33	19.63	21.83	17.60
min k3	19.78	14.96	19.49	16.85
min k5	17.06	13.52	17.57	15.55

Uniform noise — MSE, lower is better

Filter	A: uniform -0.1 0.1	A: uniform -0.3 0.3	B: uniform -0.1 0.1	B: uniform -0.3 0.3
adaptive local noise reduction k3	62.38	506.84	84.96	514.78
adaptive local noise reduction k5	61.78	467.16	85.79	481.83
adaptive median Smax7	133.76	924.52	156.95	853.59
alpha trimmed k3 d2	65.82	291.30	143.92	370.14

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alpha trimmed k3 d6	76.13	383.11	148.38	410.94
alpha trimmed k5 d2	102.34	226.80	217.60	376.68
alpha trimmed k5 d6	95.66	210.53	214.61	340.98
arithmetic mean k3	65.01	279.89	147.88	395.03
arithmetic mean k5	107.11	241.91	221.15	403.33
contraharmonic k3 Q-1	242.42	891.99	401.70	880.74
contraharmonic k3 Q1	177.09	1449.86	382.86	2032.04
contraharmonic k5 Q-1	524.36	1342.42	636.77	1208.58
contraharmonic k5 Q1	293.46	1580.36	543.23	2240.28
geometric mean k3	178.63	706.73	315.74	738.88
geometric mean k5	324.38	940.69	473.21	945.69
harmonic mean k3	242.42	891.99	401.70	880.74
harmonic mean k5	524.36	1342.42	636.77	1208.58
max k3	798.68	4151.86	1157.06	4555.30
max k5	1576.33	6039.65	2338.05	6765.00
median k3	87.54	466.95	160.87	480.91
median k5	92.25	280.60	226.83	362.46
midpoint k3	98.05	458.38	224.49	744.06
midpoint k5	239.86	708.62	426.51	1130.55
min k3	684.65	2073.40	731.31	1344.56
min k5	1278.80	2892.72	1137.59	1811.20

Salt-and-Pepper noise — PSNR (dB), higher is better

Filter	A: salt pepper d0.05	A: salt pepper d0.15	B: salt pepper d0.05	B: salt pepper d0.15
adaptive local noise reduction k3	20.13	17.57	19.04	16.98
adaptive local noise reduction k5	23.17	19.83	21.95	19.05
adaptive median Smax7	36.99	35.27	31.08	30.12
alpha trimmed k3 d2	29.95	23.51	26.30	22.16
alpha trimmed k3 d6	32.82	29.55	27.42	26.08
alpha trimmed k5 d2	27.43	23.35	24.45	21.74
alpha trimmed k5 d6	28.75	26.64	24.98	24.03
arithmetic mean k3	25.15	20.17	23.25	19.08
arithmetic mean k5	25.83	21.76	23.45	20.27
contraharmonic k3 Q-1	15.63	11.92	18.58	15.64
contraharmonic k3 Q1	12.68	8.25	10.10	6.05
contraharmonic k5 Q-1	12.09	9.67	15.58	13.60
contraharmonic k5 Q1	12.00	7.62	8.43	4.86
geometric mean k3	16.61	12.71	19.54	16.39
geometric mean k5	15.97	11.99	18.63	15.62
harmonic mean k3	15.63	11.92	18.58	15.64
harmonic mean k5	12.09	9.67	15.58	13.60
max k3	8.76	4.96	7.55	3.91
max k5	5.20	2.66	4.03	1.62
median k3	32.93	29.97	27.34	26.22
median k5	29.89	29.39	24.90	24.74
midpoint k3	14.59	11.14	13.71	10.23
midpoint k5	11.50	9.42	10.44	8.18
min k3	15.19	11.81	17.75	15.36
min k5	11.79	9.62	15.10	13.50

Salt-and-Pepper noise — MSE, lower is better

Filter	A: salt pepper d0.05	A: salt pepper d0.15	B: salt pepper d0.05	B: salt pepper d0.15
adaptive local noise reduction k3	630.53	1138.62	811.22	1302.77
adaptive local noise reduction k5	313.07	676.65	414.61	809.83
adaptive median Smax7	12.99	19.31	50.71	63.23
alpha trimmed k3 d2	65.71	289.82	152.30	395.31
alpha trimmed k3 d6	33.99	72.15	117.88	160.45
alpha trimmed k5 d2	117.61	300.34	233.18	435.62
alpha trimmed k5 d6	86.79	140.97	206.62	257.21
arithmetic mean k3	198.47	624.82	307.74	802.95
arithmetic mean k5	169.71	433.65	293.62	610.59
contraharmonic k3 Q-1	1780.12	4176.19	901.38	1773.58
contraharmonic k3 Q1	3507.60	9723.28	6359.93	16160.81
contraharmonic k5 Q-1	4019.82	7014.97	1797.48	2840.96
contraharmonic k5 Q1	4106.86	11245.37	9343.50	21221.54
geometric mean k3	1417.76	3486.85	722.75	1493.55
geometric mean k5	1644.47	4114.93	891.04	1782.29
harmonic mean k3	1780.12	4176.19	901.38	1773.58
harmonic mean k5	4019.82	7014.97	1797.48	2840.96
max k3	8654.89	20774.38	11429.04	26443.87
max k5	19639.52	35223.51	25711.53	44738.39

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median k3	33.15	65.49	120.06	155.33
median k5	66.66	74.83	210.52	218.33
midpoint k3	2258.88	4999.45	2765.32	6165.51
midpoint k5	4607.48	7427.45	5876.48	9876.70
min k3	1966.49	4284.69	1091.35	1892.75
min k5	4303.48	7095.11	2010.39	2905.66

Zoomed-in Crop — Edge/Detail Preservation

A high-detail region was located automatically (highest local Laplacian variance) in each image's salt-and-pepper ($d=0.15$) noisy input, then zoomed 4x for comparison across filters:

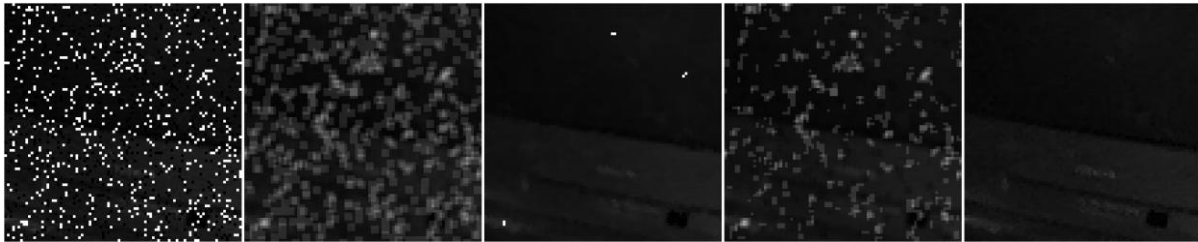


Image A (smooth) crop, left to right: noisy input, arithmetic mean $k=3$, median $k=3$, alpha-trimmed $k=3$ $d=2$, adaptive median $S_{max}=7$



Image B (textured) crop, left to right: noisy input, arithmetic mean $k=3$, median $k=3$, alpha-trimmed $k=3$ $d=2$, adaptive median $S_{max}=7$

Question - Does the filter with the best PSNR value always look the best visually?

Not always, and three pairings from the results above illustrate both agreement and disagreement between PSNR and visual quality:

- 1) **Agreement:** Median filtering outperforms the correctly signed contraharmonic filter ($Q = -1$) on salt-and-pepper noise, with both higher PSNR and better visual quality.
- 2) **Disagreement:** Alpha-trimmed filtering ($d = 2$) can achieve competitive PSNR while leaving visible speckles, whereas the median filter looks cleaner. PSNR may overlook a few strong local artifacts..
- 3) **Disagreement:** Geometric and arithmetic mean filters on Gaussian noise can have a large PSNR difference, even though their visual outputs may appear quite similar.

Follow-up - Smooth vs. Textured Sensitivity to Filter Choice

Comparing the PSNR spread across all filters for a fixed noise setting between Image A (smooth) and Image B (textured) in the Task 3 tables above (e.g. Gaussian, $\text{var}=0.05$, or salt-and-pepper $d=0.15$): the gap between the best- and worst-performing filter is the key indicator of how much filter choice matters for that image. A smooth image's flat regions are easy for almost any filter to restore reasonably well, so a poorly-chosen filter's failure to preserve genuine detail costs less there; a textured image cannot hide behind flat regions, so the gap between a well-chosen and a poorly-chosen filter tends to be more consequential on it. This matters for real-world restoration pipelines: a filter tuned and validated on flat calibration imagery may under-perform once deployed on genuinely detailed real-world content, which is exactly the gap a textured test image is meant to expose before deployment.